ecbench report: ECBS1hdf1473acacd3

Label: chain-free Koblitz IC: m = 2 half-dimension base against the m = 3 one-link chain, K_0 / F_2^23

Description: LEADS.md L2, the chained comparison. At n = 23 (the largest size the inherited-F4 m = 3 oracle has completed before, research/pdp_root_finder_ecbench_20261013) the m = 2 pair-table arms at dimension 8, 10, 12 are run against the m = 3 one-link chain at dimension 8 and 10 with the inherited-F4 oracle, the strong rho reference and its A/A control. Counted operations; every phase charged; a measurement, not a performance claim.

Generated by `ecbench report` (ecbench.report/v1). Observations only; derived numbers carry the key of their derivation (last section).

Session

field	value
session id	ECBS1hdf1473acacd3
spec id	ECS1he1fd1ce41dc7
commit	212b120dda1fa43de9b2ab75784d73c53ea68c36 (dirty)
status	complete
run options	cpus="none", lock=/tmp/crypto-bench.lock, wait_for_lock=false, allow_busy=true

Host and isolation

Intel(R) Xeon(R) Processor @ 2.10GHz; 4 logical / 4 physical cores; kernel 6.18.44-fc-v139; 15.7 GiB; env class ECBENV2he7f30f001a0e; virtualization cpu_flag=hypervisor, systemd_detect_virt_vm=kvm.

Isolation earned by the 56 measured records: L0 56, L1 0, L2 0, L3 0. Required of a wall-clock figure: L0.

blocker	measured records
L1: no CPU reservation (–cpus none, or no affinity control on this OS)	56

Arms

arm	role	method	params	expected S [d4]
rho-strong	reference	rho.signed_frobenius_strong	dp_bits=4; lanes=32; step_cap_factor=2000	0.1848
rho-strong-aa	control	rho.signed_frobenius_strong	dp_bits=4; lanes=32; step_cap_factor=2000	0.1848
ic-m2-d8	candidate	ic.pipeline	factor_base=koblitz-standard-subspace:dimension=8; linalg=incremental-gauss; max_trials=2000000; oracle=pdp3-koblitz:m=2,engine=rootfind-negfold; solver=; solver_budget_seconds=0; targets=walk	–
ic-m2-d10	candidate	ic.pipeline	factor_base=koblitz-standard-subspace:dimension=10; linalg=incremental-gauss; max_trials=2000000; oracle=pdp3-koblitz:m=2,engine=rootfind-negfold; solver=; solver_budget_seconds=0; targets=walk	–
ic-m2-d12	candidate	ic.pipeline	factor_base=koblitz-standard-subspace:dimension=12; linalg=incremental-gauss; max_trials=2000000; oracle=pdp3-koblitz:m=2,engine=rootfind-negfold; solver=; solver_budget_seconds=0; targets=walk	–
ic-m3-d8-f4	candidate	ic.pipeline	factor_base=koblitz-standard-subspace:dimension=8; linalg=incremental-gauss; max_trials=200000; oracle=pdp3-koblitz:m=3,engine=inherited-f4,degree=3,node_budget=8192; solver=; solver_budget_seconds=0; targets=walk	–
ic-m3-d10-f4	candidate	ic.pipeline	factor_base=koblitz-standard-subspace:dimension=10; linalg=incremental-gauss; max_trials=200000; oracle=pdp3-koblitz:m=3,engine=inherited-f4,degree=3,node_budget=8192; solver=; solver_budget_seconds=0; targets=walk	–

Workloads

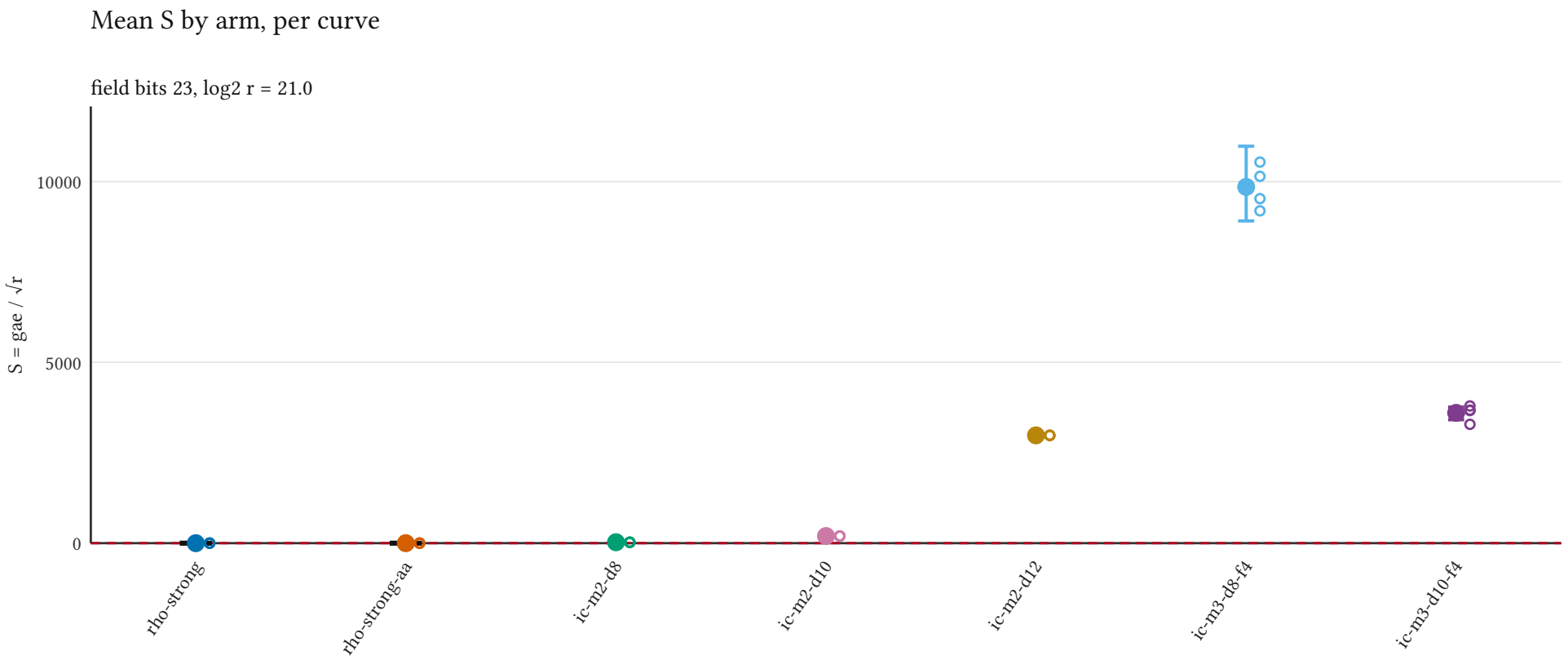
workload	curve	family	log2 r	A	target index	target seed
Wb10162dbed57	icv1-f2m23-t5197-69e76b73	koblitz	20.999	46	0	20261002
W2d1f6f595676	icv1-f2m23-t5197-69e76b73	koblitz	20.999	46	1	20261002
W33c08f0f5762	icv1-f2m23-t5197-69e76b73	koblitz	20.999	46	2	20261002
W62a208170009	icv1-f2m23-t5197-69e76b73	koblitz	20.999	46	3	20261002

Runs

Planned 84 (28 warm-up, 56 measured); records 84 (56 measured); verified 56 (56 with matches_target); not verified 0; planned without a record 0.

No record has a status other than verified.

Graphs



— dot: mean S of the arm's verified runs on the curve; whisker: 95% bootstrap interval [d2]

- - dashed red: floor √(π/2A) [d3]

— black tick: expected S [d4]; hollow dot: per-workload mean

Session ECBS1hdf1473acacd3, n = 56 verified runs. Unit: S = gae / √r.

Figure 1: Mean S by arm per curve with 95% bootstrap whiskers, the floor and expected S; session ECBS1hdf1473acacd3, n = 56 verified runs, unit gae (S = gae / √r)

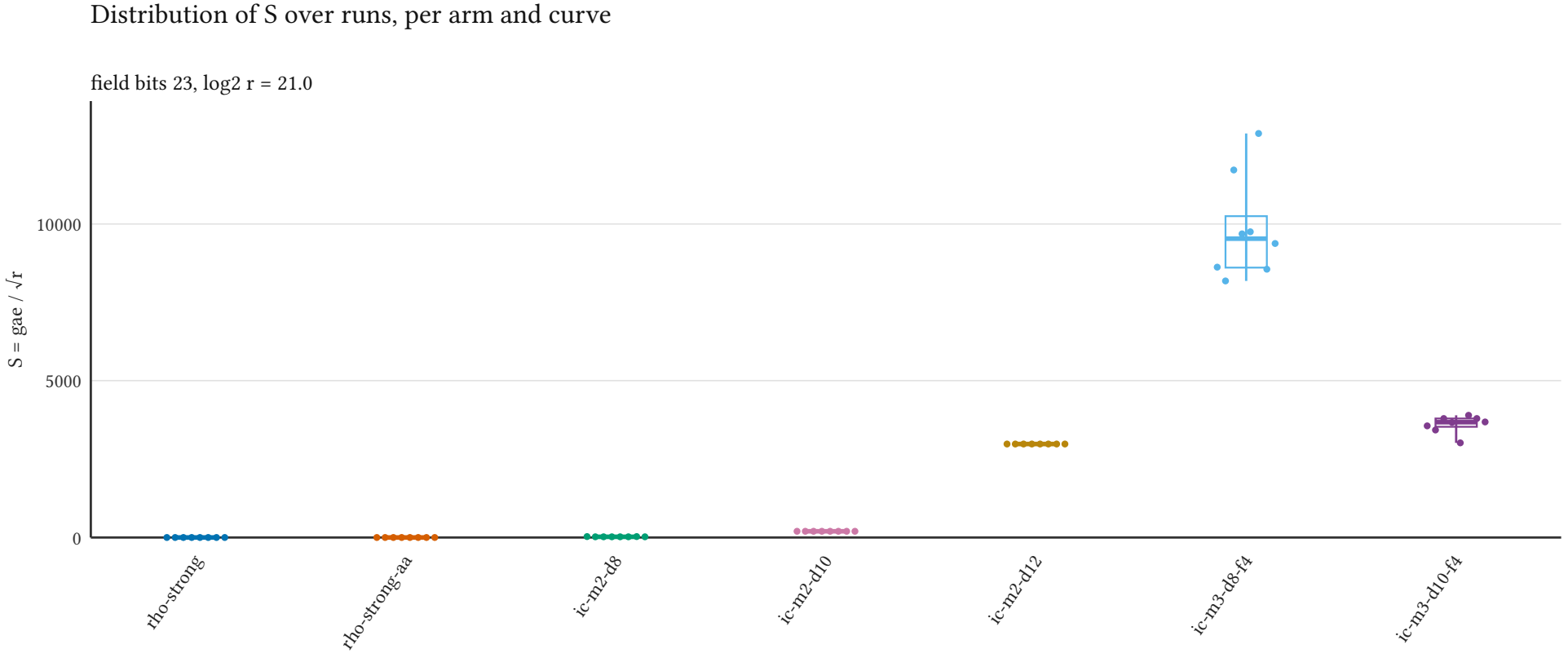
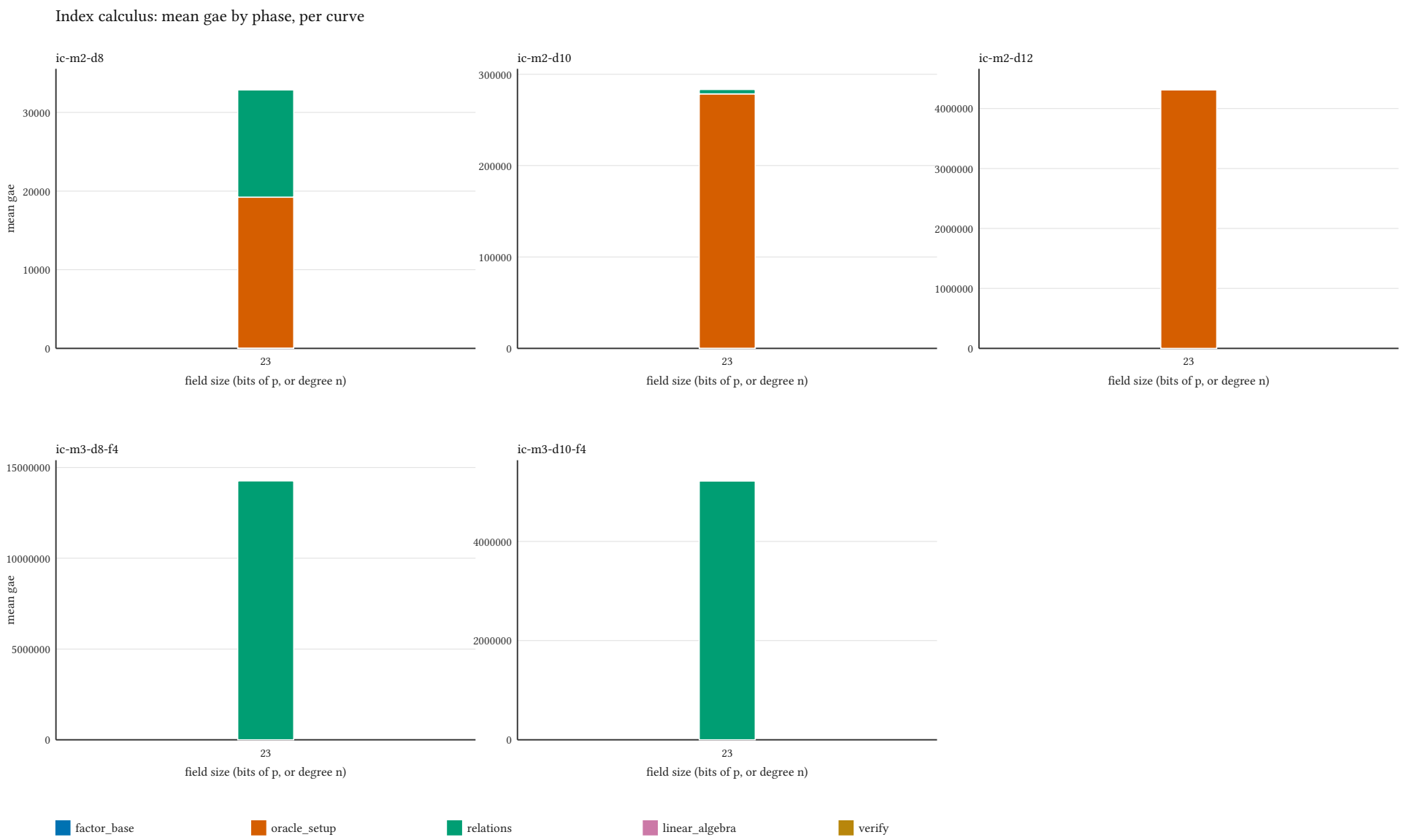


Figure 2: Distribution of S over the verified runs, per arm and curve; session ECBS1hdf1473acacd3, n = 56 verified runs, unit gae (S = gae / √r)



Session ECBS1hdf1473acacd3, n = 56 verified runs. Unit: gae. Shares [d8].

Figure 3: Mean gae by phase, index-calculus arms; session ECBS1hdf1473acacd3, n = 56 verified runs, unit gae (S = gae / √r)

Results per arm and curve

arm	curve	log2 r	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-strong	icv1-f2m23-t5197-69e76b73	20.999	8 / 8	1.3179	1.3815	1.1159	1.4032	0.1150	[1.2405, 1.3850]	0.1848	0.1848
rho-strong-aa	icv1-f2m23-t5197-69e76b73	20.999	8 / 8	1.3179	1.3815	1.1159	1.4032	0.1150	[1.2405, 1.3850]	0.1848	0.1848
ic-m2-d8	icv1-f2m23-t5197-69e76b73	20.999	8 / 8	22.7314	23.1795	19.3515	26.5242	2.8708	[20.8849, 24.5440]	0.1848	–
ic-m2-d10	icv1-f2m23-t5197-69e76b73	20.999	8 / 8	195.6623	195.6023	194.1814	196.5013	0.7267	[195.1391, 196.0890]	0.1848	–
ic-m2-d12	icv1-f2m23-t5197-69e76b73	20.999	8 / 8	2980.9944	2981.1435	2980.2281	2981.3107	0.3811	[2980.7249, 2981.2074]	0.1848	–
ic-m3-d8-f4	icv1-f2m23-t5197-69e76b73	20.999	8 / 8	9849.3293	9530.3491	8183.4627	12887.9022	1645.2833	[8912.2152, 10978.6379]	0.1848	–
ic-m3-d10-f4	icv1-f2m23-t5197-69e76b73	20.999	8 / 8	3605.5520	3677.5110	3014.6998	3899.1415	280.0074	[3410.5526, 3761.2980]	0.1848	–

arm	curve	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-strong	icv1-f2m23-t5197-69e76b73	8 / 8	1908.0	7.1319	–	–	–	–	–	5.6239	35.7150	admitted (≥L0)
rho-strong-aa	icv1-f2m23-t5197-69e76b73	8 / 8	1908.0	7.1319	–	–	–	–	–	5.7008	36.2980	admitted (≥L0)
ic-m2-d8	icv1-f2m23-t5197-69e76b73	8 / 8	32908.4	123.0115	–	–	–	–	–	6.8204	46.9390	admitted (≥L0)
ic-m2-d10	icv1-f2m23-t5197-69e76b73	8 / 8	283261.6	1058.8299	–	–	–	–	–	46.1936	83.4150	admitted (≥L0)
ic-m2-d12	icv1-f2m23-t5197-69e76b73	8 / 8	4315603.9	16131.7002	–	–	–	–	–	1454.0807	1484.0850	admitted (≥L0)
ic-m3-d8-f4	icv1-f2m23-t5197-69e76b73	8 / 8	14258934.3	53299.8070	–	–	–	–	–	65337.0223	60950.5725	admitted (≥L0)
ic-m3-d10-f4	icv1-f2m23-t5197-69e76b73	8 / 8	5219779.6	19511.5035	–	–	–	–	–	18839.9507	18100.1565	admitted (≥L0)

Results per arm and workload

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-strong	Wb10162dbed57	icv1-f2m23-t5197-69e76b73	2 / 2	1.3822	1.3822	1.3818	1.3825	0.0005	[1.3818, 1.3825]	0.1848	0.1848
rho-strong	W2d1f6f595676	icv1-f2m23-t5197-69e76b73	2 / 2	1.2378	1.2378	1.1601	1.3155	0.1099	[1.1601, 1.3155]	0.1848	0.1848
rho-strong	W33c08f0f5762	icv1-f2m23-t5197-69e76b73	2 / 2	1.2595	1.2595	1.1159	1.4032	0.2032	[1.1159, 1.4032]	0.1848	0.1848
rho-strong	W62a208170009	icv1-f2m23-t5197-69e76b73	2 / 2	1.3922	1.3922	1.3811	1.4032	0.0156	[1.3811, 1.4032]	0.1848	0.1848
rho-strong-aa	Wb10162dbed57	icv1-f2m23-t5197-69e76b73	2 / 2	1.3822	1.3822	1.3818	1.3825	0.0005	[1.3818, 1.3825]	0.1848	0.1848
rho-strong-aa	W2d1f6f595676	icv1-f2m23-t5197-69e76b73	2 / 2	1.2378	1.2378	1.1601	1.3155	0.1099	[1.1601, 1.3155]	0.1848	0.1848
rho-strong-aa	W33c08f0f5762	icv1-f2m23-t5197-69e76b73	2 / 2	1.2595	1.2595	1.1159	1.4032	0.2032	[1.1159, 1.4032]	0.1848	0.1848
rho-strong-aa	W62a208170009	icv1-f2m23-t5197-69e76b73	2 / 2	1.3922	1.3922	1.3811	1.4032	0.0156	[1.3811, 1.4032]	0.1848	0.1848
ic-m2-d8	Wb10162dbed57	icv1-f2m23-t5197-69e76b73	2 / 2	24.3803	24.3803	22.6257	26.1349	2.4814	[22.6257, 26.1349]	0.1848	–
ic-m2-d8	W2d1f6f595676	icv1-f2m23-t5197-69e76b73	2 / 2	23.8527	23.8527	23.7334	23.9721	0.1688	[23.7334, 23.9721]	0.1848	–
ic-m2-d8	W33c08f0f5762	icv1-f2m23-t5197-69e76b73	2 / 2	23.1957	23.1957	19.8672	26.5242	4.7072	[19.8672, 26.5242]	0.1848	–
ic-m2-d8	W62a208170009	icv1-f2m23-t5197-69e76b73	2 / 2	19.4970	19.4970	19.3515	19.6426	0.2058	[19.3515, 19.6426]	0.1848	–
ic-m2-d10	Wb10162dbed57	icv1-f2m23-t5197-69e76b73	2 / 2	195.6023	195.6023	195.5159	195.6887	0.1222	[195.5159, 195.6887]	0.1848	–
ic-m2-d10	W2d1f6f595676	icv1-f2m23-t5197-69e76b73	2 / 2	194.8282	194.8282	194.1814	195.4750	0.9147	[194.1814, 195.4750]	0.1848	–
ic-m2-d10	W33c08f0f5762	icv1-f2m23-t5197-69e76b73	2 / 2	196.4020	196.4020	196.3026	196.5013	0.1405	[196.3026, 196.5013]	0.1848	–
ic-m2-d10	W62a208170009	icv1-f2m23-t5197-69e76b73	2 / 2	195.8168	195.8168	195.4397	196.1939	0.5333	[195.4397, 196.1939]	0.1848	–
ic-m2-d12	Wb10162dbed57	icv1-f2m23-t5197-69e76b73	2 / 2	2981.2319	2981.2319	2981.2191	2981.2446	0.0180	[2981.2191, 2981.2446]	0.1848	–
ic-m2-d12	W2d1f6f595676	icv1-f2m23-t5197-69e76b73	2 / 2	2980.8833	2980.8833	2980.5907	2981.1760	0.4139	[2980.5907, 2981.1760]	0.1848	–
ic-m2-d12	W33c08f0f5762	icv1-f2m23-t5197-69e76b73	2 / 2	2981.1930	2981.1930	2981.0754	2981.3107	0.1664	[2981.0754, 2981.3107]	0.1848	–
ic-m2-d12	W62a208170009	icv1-f2m23-t5197-69e76b73	2 / 2	2980.6695	2980.6695	2980.2281	2981.1110	0.6243	[2980.2281, 2981.1110]	0.1848	–
ic-m3-d8-f4	Wb10162dbed57	icv1-f2m23-t5197-69e76b73	2 / 2	9189.9705	9189.9705	8625.8419	9754.0992	797.7984	[8625.8419, 9754.0992]	0.1848	–
ic-m3-d8-f4	W2d1f6f595676	icv1-f2m23-t5197-69e76b73	2 / 2	10535.6824	10535.6824	8183.4627	12887.9022	3326.5410	[8183.4627, 12887.9022]	0.1848	–
ic-m3-d8-f4	W33c08f0f5762	icv1-f2m23-t5197-69e76b73	2 / 2	10141.3153	10141.3153	8557.8733	11724.7572	2239.3251	[8557.8733, 11724.7572]	0.1848	–
ic-m3-d8-f4	W62a208170009	icv1-f2m23-t5197-69e76b73	2 / 2	9530.3491	9530.3491	9378.5466	9682.1515	214.6811	[9378.5466, 9682.1515]	0.1848	–
ic-m3-d10-f4	Wb10162dbed57	icv1-f2m23-t5197-69e76b73	2 / 2	3287.1227	3287.1227	3014.6998	3559.5456	385.2642	[3014.6998, 3559.5456]	0.1848	–
ic-m3-d10-f4	W2d1f6f595676	icv1-f2m23-t5197-69e76b73	2 / 2	3664.6288	3664.6288	3430.1162	3899.1415	331.6510	[3430.1162, 3899.1415]	0.1848	–

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
ic-m3-d10-f4	W33c08f0f5762	icv1-f2m23-t5197-69e76b73	2 / 2	3792.9453	3792.9453	3791.5741	3794.3166	1.9393	[3791.5741, 3794.3166]	0.1848	–
ic-m3-d10-f4	W62a208170009	icv1-f2m23-t5197-69e76b73	2 / 2	3677.5110	3677.5110	3672.9809	3682.0411	6.4065	[3672.9809, 3682.0411]	0.1848	–

arm	workload	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-strong	Wb10162dbed57	2 / 2	2001.0	7.4796	–	–	–	–	–	9.7334	33.3390	admitted (≥L0)
rho-strong	W2d1f6f595676	2 / 2	1792.0	6.6983	–	–	–	–	–	5.8119	36.9120	admitted (≥L0)
rho-strong	W33c08f0f5762	2 / 2	1823.5	6.8161	–	–	–	–	–	5.9088	40.2030	admitted (≥L0)
rho-strong	W62a208170009	2 / 2	2015.5	7.5338	–	–	–	–	–	4.7642	34.2690	admitted (≥L0)
rho-strong-aa	Wb10162dbed57	2 / 2	2001.0	7.4796	–	–	–	–	–	11.1149	35.5120	admitted (≥L0)
rho-strong-aa	W2d1f6f595676	2 / 2	1792.0	6.6983	–	–	–	–	–	5.7008	37.8415	admitted (≥L0)
rho-strong-aa	W33c08f0f5762	2 / 2	1823.5	6.8161	–	–	–	–	–	5.8362	41.1660	admitted (≥L0)
rho-strong-aa	W62a208170009	2 / 2	2015.5	7.5338	–	–	–	–	–	5.8069	36.0340	admitted (≥L0)
ic-m2-d8	Wb10162dbed57	2 / 2	35295.5	131.9343	–	–	–	–	–	13.3155	55.4195	admitted (≥L0)
ic-m2-d8	W2d1f6f595676	2 / 2	34531.7	129.0794	–	–	–	–	–	7.5042	50.6735	admitted (≥L0)
ic-m2-d8	W33c08f0f5762	2 / 2	33580.5	125.5237	–	–	–	–	–	7.4433	51.1200	admitted (≥L0)
ic-m2-d8	W62a208170009	2 / 2	28226.0	105.5086	–	–	–	–	–	5.9815	46.9390	admitted (≥L0)
ic-m2-d10	Wb10162dbed57	2 / 2	283174.7	1058.5052	–	–	–	–	–	54.2429	93.9505	admitted (≥L0)
ic-m2-d10	W2d1f6f595676	2 / 2	282054.0	1054.3160	–	–	–	–	–	44.9823	83.5530	admitted (≥L0)
ic-m2-d10	W33c08f0f5762	2 / 2	284332.3	1062.8325	–	–	–	–	–	44.9752	82.8000	admitted (≥L0)
ic-m2-d10	W62a208170009	2 / 2	283485.2	1059.6660	–	–	–	–	–	42.3821	76.5135	admitted (≥L0)
ic-m2-d12	Wb10162dbed57	2 / 2	4315947.6	16132.9851	–	–	–	–	–	1920.3582	1647.3085	admitted (≥L0)
ic-m2-d12	W2d1f6f595676	2 / 2	4315443.0	16131.0990	–	–	–	–	–	1410.0235	1434.3345	admitted (≥L0)
ic-m2-d12	W33c08f0f5762	2 / 2	4315891.4	16132.7749	–	–	–	–	–	1477.1690	1504.9040	admitted (≥L0)
ic-m2-d12	W62a208170009	2 / 2	4315133.5	16129.9420	–	–	–	–	–	1454.0807	1484.0850	admitted (≥L0)
ic-m3-d8-f4	Wb10162dbed57	2 / 2	13304376.6	49731.6761	–	–	–	–	–	65959.0907	59011.9365	admitted (≥L0)
ic-m3-d8-f4	W2d1f6f595676	2 / 2	15252571.9	57014.0181	–	–	–	–	–	73024.1415	71408.5340	admitted (≥L0)
ic-m3-d8-f4	W33c08f0f5762	2 / 2	14681644.1	54879.8937	–	–	–	–	–	67238.9360	66270.6640	admitted (≥L0)
ic-m3-d8-f4	W62a208170009	2 / 2	13797144.7	51573.6403	–	–	–	–	–	63905.0060	62980.1425	admitted (≥L0)
ic-m3-d10-f4	Wb10162dbed57	2 / 2	4758787.6	17788.3184	–	–	–	–	–	19646.9758	16417.5995	admitted (≥L0)
ic-m3-d10-f4	W2d1f6f595676	2 / 2	5305305.6	19831.1990	–	–	–	–	–	18652.5700	17308.7775	admitted (≥L0)
ic-m3-d10-f4	W33c08f0f5762	2 / 2	5491070.1	20525.5857	–	–	–	–	–	19290.1225	18990.2095	admitted (≥L0)
ic-m3-d10-f4	W62a208170009	2 / 2	5323955.1	19900.9109	–	–	–	–	–	18389.8808	18100.1565	admitted (≥L0)

Phases of the index-calculus arms

arm	curve	phase	runs	mean gae	share [d8]
ic-m2-d8	icv1-f2m23-t5197-69e76b73	factor_base	8	37.8	0.1%
ic-m2-d8	icv1-f2m23-t5197-69e76b73	oracle_setup	8	19181.0	58.3%
ic-m2-d8	icv1-f2m23-t5197-69e76b73	relations	8	13655.2	41.5%
ic-m2-d8	icv1-f2m23-t5197-69e76b73	linear_algebra	8	4.9	0.0%
ic-m2-d8	icv1-f2m23-t5197-69e76b73	verify	8	29.5	0.1%
ic-m2-d10	icv1-f2m23-t5197-69e76b73	factor_base	8	149.9	0.1%
ic-m2-d10	icv1-f2m23-t5197-69e76b73	oracle_setup	8	278255.0	98.2%
ic-m2-d10	icv1-f2m23-t5197-69e76b73	relations	8	4807.2	1.7%
ic-m2-d10	icv1-f2m23-t5197-69e76b73	linear_algebra	8	19.9	0.0%
ic-m2-d10	icv1-f2m23-t5197-69e76b73	verify	8	29.5	0.0%
ic-m2-d12	icv1-f2m23-t5197-69e76b73	factor_base	8	598.0	0.0%

arm	curve	phase	runs	mean gae	share [d8]
ic-m2-d12	icv1-f2m23-t5197-69e76b73	oracle_setup	8	4311851.0	99.9%
ic-m2-d12	icv1-f2m23-t5197-69e76b73	relations	8	3049.0	0.1%
ic-m2-d12	icv1-f2m23-t5197-69e76b73	linear_algebra	8	76.4	0.0%
ic-m2-d12	icv1-f2m23-t5197-69e76b73	verify	8	29.5	0.0%
ic-m3-d8-f4	icv1-f2m23-t5197-69e76b73	factor_base	8	37.8	0.0%
ic-m3-d8-f4	icv1-f2m23-t5197-69e76b73	oracle_setup	8	0.0	0.0%
ic-m3-d8-f4	icv1-f2m23-t5197-69e76b73	relations	8	14258690.5	100.0%
ic-m3-d8-f4	icv1-f2m23-t5197-69e76b73	linear_algebra	8	176.6	0.0%
ic-m3-d8-f4	icv1-f2m23-t5197-69e76b73	verify	8	29.5	0.0%
ic-m3-d10-f4	icv1-f2m23-t5197-69e76b73	factor_base	8	149.9	0.0%
ic-m3-d10-f4	icv1-f2m23-t5197-69e76b73	oracle_setup	8	0.0	0.0%
ic-m3-d10-f4	icv1-f2m23-t5197-69e76b73	relations	8	5218527.5	100.0%
ic-m3-d10-f4	icv1-f2m23-t5197-69e76b73	linear_algebra	8	1072.8	0.0%
ic-m3-d10-f4	icv1-f2m23-t5197-69e76b73	verify	8	29.5	0.0%

Raw records

`research\/koblitz\_halfdim\_base\_20261018\/sessions\/n23/records.jsonl` (84 records; ids in `raw_records.csv` and `report.json`); child stderr in `research\/koblitz\_halfdim\_base\_20261018\/sessions\/n23/exec/` (0 file(s)).

Derivations

- **[d1]** S = cost.s of each verified, non-warm-up record (`total_gae / √r`, stored by the runner); mean, median, min, max and sample stdev (stats.rs) are over those records.
- **[d2]** 95% percentile bootstrap interval of the mean S: 4000 resamples of the verified runs of the cell with replacement (stats::bootstrap_ci, seed 20261002). A cell with one run has no interval. Not a significance test.
- **[d3]** total gae and S / floor are means over the verified runs of the cell of cost.total_gae and boundaries.ratio_to_floor; floor S is boundaries.floor_s = √(π / 2A), A = automorphisms the curve offers a generic method.
- **[d4]** expected S is methods::expected_s(method, A): the closed form the method's mean S approaches as r grows (set-up omitted). A reference mark, not a measurement.
- **[d5]** scaling: per arm, per curve, the mean gae over the arm's verified runs on that curve; the line is ordinary least squares of log2(mean gae) on log2(r) over the curves (bounds::ols), slope α, intercept log2 C so gae = C·r^α. The standard error of α is √(SS_res/(sizes−2)/S_xx) from those residuals (none below 3 sizes). Counted operations, one point per size: it is not the cluster-bootstrap interval of `ecbench bound fit`, which resamples runs and sizes.
- **[d6]** counters (distinguished points, steps) and field operations (muls, sqrs, invs) are means over the cell's verified runs of the counters the record carries (steps = `steps`, or baby + giant steps for BSGS); a dash means the method did not count it (unknown, not zero).
- **[d7]** wall and CPU are medians over the cell's verified runs of time.solve_wall_ns and time.user_ns + sys_ns. The figure is admitted only when every run behind it earned the level the spec requires (compare.rs's gate); otherwise it is informative at the lowest level earned, and never a result.
- **[d8]** phase share = the phase's mean gae over the sum of the arm's mean gae across phases on that curve, over verified runs.